

ments the ATMEL application note on USB generic human interface device (HID). It uses functions findHidDevice, readData, writeData and closeDevice defined in ATUSBHID.DLL. As shown in the GUI screenshot, the software connects to the DAQ board through USB vendor ID for ATMEL (03EB in hexadecimal) and product ID for the microcontroller (204F in hexadecimal).

An 8-byte send-buffer is prepared based on the settings of digital input/output selections, their corresponding values and analogue output values. This is sent continuously through a timer event in C# routine, within an infinite loop in LabView scheme.

Similarly, the incoming eight bytes are read to obtain the six 8-bit analogue input values corresponding to the voltage available at ADC pins and two 8-bit values for the input pin status. The graphical user interface (GUI) shows the various ADC inputs and DAC outputs.

Fig. 7 shows the front-end GUI screenshot of C# program output. Fig. 8 shows the block diagram for LabView VI, which generates a GUI analogous to that generated by the C# program. Additionally, it explores the digital input and output capability as well. LabView being a block-diagram-based programming language, it explains the functional implementation of the software very clearly. A 30-day evaluation version of the latest LabView version can be freely downloaded from the Internet.

The C# program and the LabView VI are used only to demonstrate the correct functioning of the board. On similar lines, you can develop your own code in various other programming languages. (Note that LabView was not tested at EFY Lab.)

Testing and test results

Before getting started, using a multi-meter you need to measure the USB voltage that your system is supplying. Save this value in the parameter V_Max_USB of the program.

The developed prototype was tested with various sensors like po-

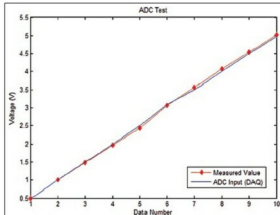


Fig. 9: Plot of ADC input voltage read by the DAQ device shown in software vs value measured using multimeter

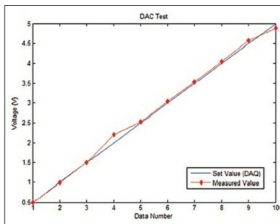


Fig. 10: Plot of DAC output voltage set in software vs value measured using multimeter

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

tentiometer, LDR and analogue accelerometer. The sensed ADC output shown by the software was found to be closely matching with the physical measurements.

Similarly, the DAC outputs were tested physically using a digital multi-meter and found to be in agreement with the desired voltage values set in the software. Maximum variation from the actual value was found to be within $\pm 0.02V$. Apart from the ADC and DAC tests, the digital inputs and outputs were found to be responding properly to the software. Figs 9 and 10 show the

plots for the ADC and DAC tests, respectively.

With Generic HID mode of USB communication, the device runs at 9.6 kilo-samples per second, which is good enough to handle a situation where the inputs and outputs are dynamically varying. Though the developed DAQ device is quite basic compared to the existing technology, it can be very helpful for everyday use where one needs an inexpensive device with reasonably good reliability and precision. It has the potential to replace advanced DAQ devices like NI USB 6008/6009 (which it closely resembles to) in smaller enterprises that don't have much stringent demands and certification requirements.

Troubleshooting

If the software on your PC doesn't respond, first check the circuit for proper connections. You can also try closing the appli-

cation and restarting it. If the problem persists, check voltages at various test points as per Table II.

The author has uploaded on YouTube videos of this DAQ device being tested using C# and LabView interface. You may watch the same for better understanding.

Further application

The board can be reprogrammed for different purposes with different customised programs for the connected I/Os. ●

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